

SAFETY DATA SHEET

NALMET® 1689

Section: 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : NALMET® 1689

Other means of identification : Not applicable.

Recommended use : WATER CLARIFICATION AID

Restrictions on use : Refer to available product literature or ask your local Sales Representative for restrictions on use and dose limits.

Company : Nalco Middle East FZE
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Issuing date : 12/21/2022

Section: 2. HAZARDS IDENTIFICATION

GHS Classification

Corrosive to metals : Category 1
Eye irritation : Category 2B

GHS Label element

Hazard pictograms :



Signal Word : Warning

Hazard Statements : May be corrosive to metals.
Causes eye irritation.

Precautionary Statements : **Prevention:**
Keep only in original container. Wash skin thoroughly after handling.
Response:
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Absorb spillage to prevent material damage.
Storage:

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Store in corrosive resistant container with a resistant inner liner.

Other hazards : None known.

Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Chemical Name	CAS-No.	Concentration: (%)
Sodium Chloride	7647-14-5	1 - 5
Sodium Sulphide	1313-82-2	1 - 5
Sodium Hydroxide	1310-73-2	0.1 - 1

Section: 4. FIRST AID MEASURES

- In case of eye contact : Rinse with plenty of water. Get medical attention if symptoms occur.
- In case of skin contact : Wash off with soap and plenty of water. Get medical attention if symptoms occur.
- If swallowed : Rinse mouth. Get medical attention if symptoms occur.
- If inhaled : Get medical attention if symptoms occur.
- Protection of first-aiders : In event of emergency assess the danger before taking action. Do not put yourself at risk of injury. If in doubt, contact emergency responders. Use personal protective equipment as required.
- Notes to physician : Treat symptomatically.
- Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

Section: 5. FIREFIGHTING MEASURES

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Specific hazards during firefighting : Not flammable or combustible.
- Hazardous combustion products : Decomposition products may include the following materials: Carbon oxides nitrogen oxides (NOx) Sulphur oxides metal oxides Hydrogen chloride
- Special protective equipment for firefighters : Use personal protective equipment.
- Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

Section: 6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

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- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway. Flush away traces with water.

Section: 7. HANDLING AND STORAGE

- Advice on safe handling : Wash hands thoroughly after handling. Use only with adequate ventilation.
- Conditions for safe storage : Do not store near acids. Keep out of reach of children. Keep container tightly closed. Store in suitable labelled containers. Keep only in original packaging. Absorb spillage to prevent material damage.
- Suitable material : The following compatibility data is suggested based on similar product data and/or industry experience: PVC, EPDM, Buna-N, HDPE (high density polyethylene), Polyurethane, Hypalon, Viton, Neoprene, Polypropylene, Polyethylene, Compatibility with Plastic Materials can vary; we therefore recommend that compatibility is tested prior to use., Stainless Steel 316L
- Unsuitable material : The following compatibility data is suggested based on similar product data and/or industry experience: Brass, Mild steel, 100% phenolic resin liner, Stainless Steel 304

Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
Sodium Hydroxide	1310-73-2	C	2 mg/m ³	ACGIH
		C	2 mg/m ³	NIOSH REL
		TWA	2 mg/m ³	OSHA Z-1
		C	2 mg/m ³	OSHA P0
		C	2 mg/m ³	
Sodium Hydroxide	1310-73-2	Ceiling limit	2 mg/m ³	ARE OEL
Sodium Hydroxide	1310-73-2	CLV	2 mg/m ³	BH OEL

- Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

- Eye protection : Safety glasses
- Hand protection : Wear protective gloves.
- Skin protection : Wear suitable protective clothing.
- Respiratory protection : When respiratory risks cannot be avoided or sufficiently limited by technical means of collective protection or by measures, methods or procedures of work organization, consider the use of certified respiratory protection equipment meeting EU requirements (89/656/EEC, (EU) 2016/425), or equivalent, with filter type:

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Filter type: P, Particulates type

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling.

The Personal Protective Equipment (PPE) recommendations provided above have been made in good faith based on typical expected conditions of use. PPE selection should always be completed in conjunction with a proper risk assessment and in accordance with a PPE management program.

Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Liquid
Colour	: brown
Odour	: Sulfurous
Flash point	: , Method: ASTM D 93, Pensky-Martens closed cup, does not flash
pH	: 13.1,(100 %)
Odour Threshold	: no data available
Melting point/freezing point	: -15 °C
Initial boiling point and boiling range	: 108 °C
Evaporation rate	: no data available
Flammability (solid, gas)	: Not applicable.
Upper explosion limit	: Not applicable.
Lower explosion limit	: Not applicable.
Vapour pressure	: 2.6 kPa, (25 °C),
Relative vapour density	: no data available
Relative density	: 1.225, (25 °C),
Density	: 10.18 lb/gal
Water solubility	: completely soluble
Solubility in other solvents	: no data available
Partition coefficient: n-octanol/water	: no data available
Auto-ignition temperature	: no data available
Thermal decomposition	: no data available
Viscosity, dynamic	: no data available
Viscosity, kinematic	: no data available
Molecular weight	: no data available
VOC	: 0 %, 0 g/l

Section: 10. STABILITY AND REACTIVITY

Reactivity	: No dangerous reaction known under conditions of normal use.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous	: No dangerous reaction known under conditions of normal use.

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reactions

Conditions to avoid	: None known.
Incompatible materials	: Strong acids Aluminium Mild steel
Hazardous decomposition products	: In case of fire, hazardous decomposition products may be produced such as: Carbon oxides nitrogen oxides (NOx) Sulphur oxides metal oxides Hydrogen chloride

Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Eyes	: Causes eye irritation.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Eye contact	: Redness, Irritation
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

Toxicity

Product

Acute oral toxicity	: Acute toxicity estimate: > 5,000 mg/kg
Acute inhalation toxicity	: no data available
Acute dermal toxicity	: no data available
Skin corrosion/irritation	: no data available
Serious eye damage/eye irritation	: no data available
Respiratory or skin sensitization	: no data available
Carcinogenicity	: no data available
Reproductive effects	: no data available

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Germ cell mutagenicity : no data available
Teratogenicity : no data available
STOT - single exposure : no data available
STOT - repeated exposure : no data available
Aspiration toxicity : no data available

Components

Acute dermal toxicity : Sodium Chloride
LD50 Rat: > 10,000 mg/kg

Section: 12. ECOLOGICAL INFORMATION

Toxicity

Environmental Effects : Harmful to aquatic life.

Product

Toxicity to fish : LC50 Oncorhynchus mykiss (rainbow trout): 74 mg/l
Exposure time: 96 hrs
Test substance: Product

LC50 Cyprinodon variegatus (sheepshead minnow): > 1,000 mg/l
Exposure time: 96 hrs
Test substance: Product

NOEC Oncorhynchus mykiss (rainbow trout): < 40 mg/l
Exposure time: 96 hrs
Test substance: Product

NOEC Cyprinodon variegatus (sheepshead minnow): 400 mg/l
Exposure time: 96 hrs
Test substance: Product

NOEC Pimephales promelas (fathead minnow): 432 mg/l
Exposure time: 96 h
Test substance: Product

LC50 Pimephales promelas (fathead minnow): 602 mg/l
Exposure time: 96 h
Test substance: Product

Toxicity to daphnia and other aquatic invertebrates : LC50 Daphnia magna (Water flea): 73 mg/l
Exposure time: 48 hrs
Test substance: Product

EC50 Daphnia magna (Water flea): 18 mg/l
Exposure time: 48 hrs
Test substance: Product

NOEC Daphnia magna (Water flea): 5 mg/l
Exposure time: 48 hrs
Test substance: Product

Toxicity to fish (Chronic) : ChV: 85.6 mg/l

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toxicity)

Exposure time: 7 d
Species: Fathead Minnow
Test substance: Product

LOEC: 121 mg/l
Exposure time: 7 d
Species: Fathead Minnow
Test substance: Product

NOEC: 60.5 mg/l
Exposure time: 7 d
Species: Fathead Minnow
Test substance: Product

EC25 / IC25: 27.2 mg/l
Exposure time: 7 d
Species: Fathead Minnow
Test substance: Product

Chronic Toxicity Value: 10.7 mg/l
Exposure time: 7 d
Species: Fathead Minnow
Test substance: Product

LOEC: 15.1 mg/l
Exposure time: 7 d
Species: Fathead Minnow
Test substance: Product

NOEC: 7.56 mg/l
Exposure time: 7 d
Species: Fathead Minnow
Test substance: Product

Toxicity to daphnia and other
aquatic invertebrates
(Chronic toxicity)

: Chronic Toxicity Value: > 4.84 mg/l
Exposure time: 7 d
Species: Ceriodaphnia dubia
Test substance: Product

LOEC: > 4.84 mg/l
Exposure time: 7 d
Species: Ceriodaphnia dubia
Test substance: Product

NOEC: 4.84 mg/l
Exposure time: 7 d
Species: Ceriodaphnia dubia
Test substance: Product

EC25 / IC25: 2.75 mg/l
Exposure time: 7 d
Species: Ceriodaphnia dubia
Test substance: Product

Chronic Toxicity Value: 3.42 mg/l
Exposure time: 7 d
Species: Ceriodaphnia dubia
Test substance: Product

LOEC: 4.84 mg/l

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Exposure time: 7 d
Species: Ceriodaphnia dubia
Test substance: Product

NOEC: 2.42 mg/l
Exposure time: 7 d
Species: Ceriodaphnia dubia
Test substance: Product

Persistence and degradability

The organic portion of this preparation is expected to be poorly biodegradable.

Chemical Oxygen Demand (COD): 420,000 mg/l

Mobility

The environmental fate was estimated using a level III fugacity model embedded in the EPI (estimation program interface) Suite TM, provided by the US EPA. The model assumes a steady state condition between the total input and output. The level III model does not require equilibrium between the defined media. The information provided is intended to give the user a general estimate of the environmental fate of this product under the defined conditions of the models.

If released into the environment this material is expected to distribute to the air, water and soil/sediment in the approximate respective percentages;

Air	: <5%
Water	: 30 - 50%
Soil	: > 90%

The portion in water is expected to be soluble or dispersible.

Bioaccumulative potential

This preparation or material is not expected to bioaccumulate.

Other information

no data available

Section: 13. DISPOSAL CONSIDERATIONS

Disposal methods : Do not contaminate storm water drains, natural waterways or soil with chemical or used container. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of contents/container in accordance with local regulations Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers.

Section: 14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (ADR)

UN number: UN 3266

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UN proper shipping name: CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S. (Sodium Sulphide)
Transport hazard class(es): 8
Packing group: III
Environmental hazards: No

Air transport (IATA)

UN number: UN 3266
UN proper shipping name: CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S. (Sodium Sulphide)
Transport hazard class(es): 8
Packing group: III
Environmental hazards: No

Sea transport (IMDG/IMO)

UN number: UN 3266
UN proper shipping name: CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S. (Sodium Sulphide)
Transport hazard class(es): 8
Packing group: III
Environmental hazards: No

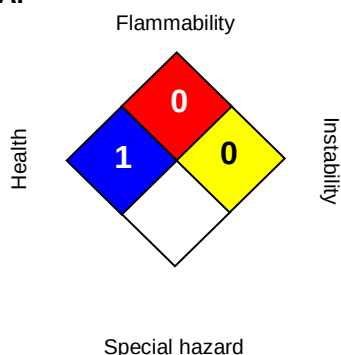
Section: 15. REGULATORY INFORMATION

NATIONAL REGULATIONS

National regulations in the country where this substance is used, should be followed.

Section: 16. OTHER INFORMATION

NFPA:



HMIS III:

HEALTH	1
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High,
4 = Extreme, * = Chronic

Revision Date : 12/21/2022
Version Number : 1.3
Prepared By : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.